

Hands-on: AWS Core Services (S3, Glue, CloudWatch, Athena)

In this hands-on, you will use **AWS** and **work with a few fundamental services to query a database**. You will create an S3 bucket, load a CSV file into S3, set up an IAM role, create a Glue crawler to crawl raw data, and use Athena to query the crawled database.

Dataset: <https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data>

Workflow of hands-on:

1. Configure Amazon S3 Buckets to load raw data and store the processed data.
2. Create an IAM role with S3 and Glue access for the crawler that will be created next.
3. Create a crawler in Glue Studio to auto-load the raw data into tables.
4. Run the Crawler and open CloudWatch to monitor its status.
5. Open Athena Editor to query the DB that has been crawled using the Glue crawler.

Queries to Run in Athena (Apply limit condition to 10 for all queries):

Query 1 — Basic Table Exploration: Write a query that retrieves the first 10 records

Query 2 — Orders by Product Category: Write a query that returns count of each product category along with the total number of orders placed in that category.

Query 3 — Revenue and Quantity by Fulfilment Method: Write a query that returns each fulfilment method with its total number of orders, total units sold, and total revenue — excluding cancelled and pending orders — sorted by highest revenue first.

Query 4 — Monthly Sales Trend: Write a query that returns each month along with the total number of orders and total revenue generated in that month, excluding canceled and pending orders, sorted chronologically from earliest to latest.

Query 5 — Top 5 Best-Selling SKUs per Category: Write a query that returns the top 5 SKUs in each product category ranked by total revenue, showing the category, SKU, total revenue, total units sold, and rank — excluding canceled, pending, and zero-quantity orders.

!!! Submission Guidelines !!!

Submit:

- Submit the SQL query script and the CSV results for each query in a GitHub repository.
- Your final README with explanations, approach, and a screenshot of CloudWatch, IAM role, and S3 Buckets.